

development boards.

This board finds a strong contender in UDOO X86 Ultra, which sports a quad-core Intel N3710 SoC clocked at 2.56GHz. Icing on the cake is Intel HD 405 graphics on board, which are supported by 8GB DDR3 RAM. Come to think of it, my PC has got poorer configuration, much poorer!

An increasing number of interface options

A lot of development boards come with a multitude of interface options ranging from the very popular USB to technical ports like RS232. T. Anand, technical architect, Knewron, says, "Interfaces like MOD-BUS, RS232 or peripheral component interconnect (PCI) slots still exist and could be prominently seen on industrial boards, while general-purpose boards usually have USB interfaces."

Arduino, for example, offers Registered Jack 45 (RJ45) and USB 2.0, among others, for connecting and loading programs from a computer. Most ARM based devices include a Joint Test Action Group (JTAG) interface. This can be used to program on-chip or external flash, or load directly into RAM. The JTAG is also the interface for on-chip debug and trace features. Wireless is also being used as one of the interfaces.

RS232 and PCI have been the norm for many years. However, RS232 seems to be an endangered port on computers today. This has brought in the need for a different port on the boards. USB is becoming more and more popular with developers in this regard. Additionally, a separate port is often required for debugging and other purposes. JTAG or single-wire debug solutions come in handy for this purpose.

It is not just the interface, though. Another reason for the extreme popularisation of development boards can be traced to the wide availability of easily-available open

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source hardware that allow a lot of customisation. However, it all started with a device that could interact with the environment using actuators and sensors to bring about a clear output that corresponded to an electrical or optical signal. Since then, hobbyists have solved many complex problems ranging from daily applications to the IoT space.

Open source hardware have captured a major chunk of the market for hobbyists. The charm of availability and vast communities for solving issues could be the major factors behind their popularity. Open source allows availability of designs, which can be tinkered with to suit project requirements—another advantage. But as the open source community grows, so do the number of open source boards.

The processing force behind the boards

Hrishikesh Kamat, chief executive officer, Shalaka Technologies, adds "ARM based processors are very common in development boards." A lot of development boards provide 32-bit solutions, hence calling for ARM Cortex-A9 processors in budget devices. Their applicability as single processors or as a scalable solution with multi-core capability make these ideal for development boards.

A9 processors have been the go-to solution for low-power thermally-constrained devices. The earlier version of A9, ARM cortex A8, still adorns BeagleBone Black, whereas Raspberry Pi 2 houses 900MHz quad-core ARM Cortex A7. Clocked at 1GHz, AM335x processor on BeagleBone provides

for low power consumption and open source hardware compatibility. So it makes sense to get familiar with ARM architecture for the long run.

8-bit AVR processor ATmega328 from Atmel is another popular development board. With specifications at 32kB flash, 2kB SRAM and 1kB EEPROM, it has a wide support community. However, in case of higher processing power, ARM Cortex-A17 is a solution to look out for. Designed to deliver high performance in a cost-efficient power and area budget, Cortex-A17 sports a 32-bit software ecosystem for mid-range consumer and embedded design. ARM Cortex processors look to be the mainline solution for Linux based development boards.

Which one to select

Choosing a development board would depend largely on the application. For a hobbyist, the applications could range from basic learning about system development to developing professional applications. Kamat, however, advises on getting to know the problem and tackling it on a step-by-step basis. He adds, "Once you know the problem, you can devise a solution and select a proper board with the required peripherals."

Deriving value from development. Anand says, "If you are interested in practising or playing, open source development boards are a good starting point." These get you to develop electronics without having to go through the hassles involved in manufacturing.

Value can also be derived from customised development boards catering to specific applications in the form of reduced timelines and speeding up product development significantly. But once the design is successful, you will have to move towards designing the circuit by weeding out unnecessary parts and bringing in the required components. **EFY**